

CRITICAL CONDUCTORS OF SMALL ARCHIMEDEAN DATA

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ABSTRACT. For an archimedean type $\Gamma_{\mathbb{R}}^a \Gamma_{\mathbb{C}}^b$ with one simple pole, the least conductor at which the explicit formula admits a solution by positive measures on both sides is the optimum of the explicit-formula method of Odlyzko, Poitou and Serre for that type. We bracket it by computer-assisted certificates for $\Gamma_{\mathbb{R}}^2$ and $\Gamma_{\mathbb{C}}$, $4.98485094226764 \leq q_{\min} \leq 4.9994844$ and $2.981236704765 \leq q_{\min} \leq 2.997174$, and compare the computed values for seven types with the smallest discriminants of number fields with the same archimedean data. With the gap $\log 2/2\pi$ common to all data, in every case tested the genuine Dedekind zeta function lies strictly above its critical conductor, while for the data of the Riemann zeta function $1 - 7.3 \cdot 10^{-906} \leq q_{\min}$, and $q_{\min} \leq 1$ under the Riemann hypothesis. The two certified brackets hold for every gap up to $\log 4.049/2\pi$, resp. $\log 3.011/2\pi$; so the data of $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{-3})$ are not critical at their natural gaps $\log 4/2\pi$ and $\log 3/2\pi$ either. We also certify an admissible pair for the data of $\mathbb{Q}(i)$, and that the pole weight of ζ 's data, the residue of $-\zeta'/\zeta$ at $s = 1$, is pinned: no admissible pair exists for weights $r < 1 - 5.5794 \cdot 10^{-40}$ or $r > 1 + 7.8551 \cdot 10^{-21}$.

Status. This is a short computational note accompanying the companion paper [4]; it has not been peer-reviewed. Its theorems and propositions are computer-assisted certificates, each with its script and parameter file (Appendix A), and are proved. The comparison of critical conductors with minimal discriminants (Section 5), the slack as a function of the gap (Numerical observation 3.4) and the evidence on uniqueness at criticality are numerical observations: computed by finite linear programmes, not certified, and not used in any proof. The comparisons use the common gap ξ_2 . For $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{-3})$ the certified conclusions also hold at their natural gaps (Corollary 3.3); for the other types the computations were made at the gap ξ_2 only (Remark 3.6). Whether the set of admissible pairs at a critical conductor is a single pair is open for every type, ζ 's included (Question 6.1).

1. INTRODUCTION

Explicit-formula bounds for discriminants and conductors [6, 8, 3, 7] test the explicit formula of an L -function against a function $F \geq 0$ whose Fourier transform is non-negative beyond the first prime. Miller [5] used the same positivity, with carefully chosen test functions, to show that L -functions of many archimedean types have a lower first zero than ζ , or do not exist. For fixed archimedean data, the best bound that the method can give is the least conductor $q_{\min}$ at which the explicit formula admits a solution by positive measures on both sides, and by linear-programming duality it is $e^{-2\pi\kappa^*}$, where κ^* is the slack of the data at conductor 1 [4]. For the data of ζ (the factor $\Gamma_{\mathbb{R}}$ and one simple pole) the companion paper certifies $\kappa^* \leq 1.1508391 \cdot 10^{-906}$, so that $q_{\min} \geq 1 - 7.3 \cdot 10^{-906}$, while $q_{\min} \leq 1$ under the Riemann hypothesis; only the first of these two inequalities is unconditional. It also shows that the data of $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{-3})$ are not critical, with the same gap ξ_2 and also at their natural gaps, which are larger because the prime-ideal norms of these fields start at 4 and 3 (Fact 2.3). Here

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we state the same certificates for the critical conductors at every gap (Corollary 3.3) and discuss the role of the gap (Remark 3.6).

This note records the corresponding computations for other archimedean types $\Gamma_{\mathbb{R}}^a \Gamma_{\mathbb{C}}^b$ with one simple pole, in the spirit of Odlyzko's problem of determining the optimal explicit-formula bounds [7]. We certify brackets for the critical conductors of $\Gamma_{\mathbb{R}}^2$ and $\Gamma_{\mathbb{C}}$ (Theorem 3.1 and Proposition 3.2), valid for every gap up to slightly beyond the natural gaps of $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{-3})$ (Corollary 3.3); the lower ends improve the corresponding degree-2 entries of Odlyzko's tables (Remark 3.5). We certify an admissible pair for the data of $\mathbb{Q}(i)$ (Proposition 3.7) and a two-sided pin of the pole weight of ζ 's data (Theorem 4.1). For seven types, computed critical conductors are compared with the smallest discriminants of number fields with the same archimedean data (Numerical observation 5.1): with the gap ξ_2 , in every case the genuine field lies strictly above its critical conductor, and the computed critical objects are not arithmetic.

2. SETTING

We recall the definitions and results of [4] that are used; statements quoted from it are called Facts. We write $\widehat{F}(\xi) = \int_{\mathbb{R}} F(t) e^{-2\pi i \xi t} dt$, $\xi_n = \log n / (2\pi)$, and $\xi_2 = \log 2 / (2\pi)$ for the *gap*, which is the same for all archimedean data below, except in Corollary 3.3 (Remark 3.6). The test class $\mathcal{T} = \bigcup_{0 < \delta < 1/2} \mathcal{T}_{\delta}$ consists of the even functions F , real on $\mathbb{R}$ and analytic in a neighbourhood of $\{| \operatorname{Im} z | \leq \frac{1}{2} + \delta\}$ for some δ , with $\sup(1 + |z|)^2 |F(z)| < \infty$ there.

For a degree $d \geq 1$, shifts $\kappa = (\kappa_1, \dots, \kappa_d)$, a conductor $q > 0$ and a pole weight $r > 0$ put

$$\mathcal{A}_{\kappa, q, r}(F) := r(F(\tfrac{i}{2}) + F(-\tfrac{i}{2})) + \int_{\mathbb{R}} F(t) \rho(t) dt,$$

$$\rho(t) := \frac{1}{2\pi} \left[\log q + \sum_{j \leq d} (\operatorname{Re} \psi(\tfrac{\frac{1}{2} + \kappa_j + it}{2}) - \log \pi) \right],$$

with $\psi = \Gamma'/\Gamma$. The data of ζ are $d = 1$, $\kappa = (0)$, $q = r = 1$; $\Gamma_{\mathbb{R}}^2$ has $\kappa = (0, 0)$; $\Gamma_{\mathbb{C}}(s) \propto \Gamma_{\mathbb{R}}(s) \Gamma_{\mathbb{R}}(s+1)$ has $\kappa = (0, 1)$; in general $\Gamma_{\mathbb{R}}^a \Gamma_{\mathbb{C}}^b$ has $a + b$ shifts 0 and b shifts 1. The *pole weight* r is the residue at $s = 1$ of $-L'/L$ for an L -function L with these data, that is, the coefficient of $F(\tfrac{i}{2}) + F(-\tfrac{i}{2})$ in its explicit formula; it is not the residue of L itself, which changes when L is multiplied by a non-zero constant, while $-L'/L$ and the explicit formula do not. A simple pole gives $r = 1$. An *admissible pair* for $\mathcal{A}' = \mathcal{A}_{\kappa, q, r}$ is a pair (μ, ν) of positive Radon measures, μ even on $\mathbb{R}$ and ν on $[\xi_2, \infty)$, such that $\mathcal{A}'(F) = \int F d\mu + \frac{1}{\pi} \int \widehat{F} d\nu$ for every $F \in \mathcal{T}$, with absolutely convergent integrals; $\mathcal{K}(\mathcal{A}')$ is the set of admissible pairs. Put

$$\mathcal{C} := \{F \in \mathcal{T} : F \geq 0 \text{ on } \mathbb{R}, \widehat{F} \geq 0 \text{ on } [\xi_2, \infty)\}, \quad \mathcal{C}_{\text{OPS}} := \{F \in \mathcal{T} : F \geq 0 \text{ and } \widehat{F} \geq 0 \text{ on } \mathbb{R}\},$$

$\kappa^*(\mathcal{A}') := \inf\{\mathcal{A}'(F) : F \in \mathcal{C}, \int F = 1\}$, and $\kappa_{\text{OPS}}^*(\mathcal{A}')$ likewise with $\mathcal{C}_{\text{OPS}}$. Since $\mathcal{A}_{\kappa, q', r} = \mathcal{A}_{\kappa, q, r} + \frac{\log(q'/q)}{2\pi} \int$, feasibility is monotone in q , and $\kappa^*(\mathcal{A}_{\kappa, q, r}) = \kappa^*(\mathcal{A}_{\kappa, 1, r}) + \frac{\log q}{2\pi}$. The *critical conductor* is $q_{\min}(\kappa, r) := \inf\{q : \mathcal{K}(\mathcal{A}_{\kappa, q, r}) \neq \emptyset\}$.

Other gaps. For $g > 0$, replacing $[\xi_2, \infty)$ by $[g, \infty)$ in these definitions defines admissible pairs at the gap g , the cone $\mathcal{C}_g$, the slack κ_g^* and the critical conductor $q_{\min, g}$; for $g = \xi_2$ they are the objects above. A larger gap allows fewer pairs and gives a larger cone, so $q_{\min, g}$ is non-decreasing and κ_g^* is non-increasing in g .

Lemma 2.1 (Weak duality and the critical conductor). (a) *If (μ, ν) is admissible for $\mathcal{A}'$ and $F \in \mathcal{C}$, then for Borel sets $I \subseteq \mathbb{R}$ and $J \subseteq [\xi_2, \infty)$, $(\inf_I F) \mu(I) + \frac{1}{\pi} (\inf_J \widehat{F}) \nu(J) \leq \mathcal{A}'(F)$; in particular $\mathcal{A}'(F) \geq 0$, and if $\int F = 1$ no admissible μ satisfies $\mu \geq \lambda dt$ with $\lambda > \mathcal{A}'(F)$.* (b) *An admissible pair exists if and only if $\kappa^*(\mathcal{A}') \geq 0$; hence $q_{\min}(\kappa, r) = e^{-2\pi \kappa^*(\mathcal{A}_{\kappa, 1, r})}$.*

Proof. For ζ 's data, (a) is [4, Lemma 2.5] and (b) is [4, Theorem 2.7 and Proposition 2.10]. Part (a) needs one line: for an admissible pair and $F \in \mathcal{C}$, both integrands in $\mathcal{A}'(F) = \int F d\mu + \frac{1}{\pi} \int \widehat{F} d\nu$ are non-negative where the measures live. For (b), the proofs in [4] use only that the archimedean functional is $\|\cdot\|_\delta$ -continuous on each $\mathcal{T}_\delta$ and the bounds for its values on the test functions of the proof of [4, Lemma 2.6], which follow from the growth of the archimedean weight. Both hold for $\mathcal{A}_{\kappa,q,r}$: its pole term $r(F(\frac{i}{2}) + F(-\frac{i}{2}))$ is $\|\cdot\|_\delta$ -continuous, and ρ is continuous with $\rho(t) = \frac{d}{2\pi} \log|t| + O(1)$, as for ζ with $d = 1$. So these proofs apply verbatim; the conductor enters as $\frac{\log q}{2\pi}$ times Lebesgue measure, which gives the formula for $q_{\min}$. $\square$

Lemma 2.2 (Herglotz form of an admissible pair). *Let $\tilde{\nu} \geq 0$ be a measure on $[2, \infty)$ such that $\sigma := \tilde{\nu} - r\mathbf{1}_{[2, \infty)}(x) dx$ satisfies $\int x^{-1/2} d|\sigma|(x) < \infty$, and put $\mathcal{L}(s) := \int x^{-s} d\tilde{\nu}(x)$ for $\operatorname{Re} s > 1$. Then $\mathcal{L}$ extends continuously to $\operatorname{Re} s \geq \frac{1}{2}$, $s \neq 1$, as $r2^{1-s}/(s-1) + \int x^{-s} d\sigma$. Let $\mu(t) := \rho(t) - \frac{1}{\pi} \operatorname{Re} \mathcal{L}(\frac{1}{2} + it)$, and let ν be the image of $x^{-1/2} \tilde{\nu}$ under $x \mapsto \log x/2\pi$. Then $\mathcal{A}_{\kappa,q,r}(F) = \int F d\mu + \frac{1}{\pi} \int \widehat{F} d\nu$ for every $F \in \mathcal{T}$. Hence $(\mu dt, \nu)$ is admissible for $\mathcal{A}_{\kappa,q,r}$ if and only if $\mu \geq 0$.*

Proof. For $r = 1$ and the data $\Gamma_{\mathbb{R}}, \Gamma_{\mathbb{R}}^2, \Gamma_{\mathbb{C}}$ this is [4, Lemma 5.7]. Its proof shifts the contour in $\frac{1}{\pi} \int \widehat{F} d\nu = \frac{1}{\pi} \int F(w) \mathcal{L}(\frac{1}{2} + iw) dw$ past the simple pole of $\mathcal{L}$ at $s = 1$, whose residue, here r , produces the pole term $r(F(\frac{i}{2}) + F(-\frac{i}{2}))$, while the archimedean weight enters only through the definition of μ . So the proof applies verbatim for every $r > 0$ and every κ . $\square$

Explicit pairs. The pairs certified below have the following form: on $x \geq 1$,

$$\tilde{\nu} = \mathbf{1}_{[2, \infty)}(x) dx - \sum_{i < n} f_i \mathbf{1}_{[e^{v_i}, e^{v_{i+1}}]}(x) dx + \sum_b (\text{bump}_b) - \sum_m a_m x^{-m} \mathbf{1}_{(X, \infty)}(x) dx,$$

with $0 \leq f_i \leq 1$, cell edges $\log 2 = v_0 < \dots < v_n$, $X = e^{v_n}$, $a_m \geq 0$, and bumps $w_b x^{1/2} e^{-u_b/2} \phi(\log x - u_b) d(\log x)$ with $w_b \geq 0$, where ϕ is the density of a rescaled Irwin–Hall distribution, supported on $[-\varepsilon, \varepsilon]$, with Fourier transform $S(t) = \operatorname{sinc}(\varepsilon t/k)^k$ (for $\varepsilon = 0$, atoms). Then $\tilde{\nu} \geq 0$ with support in $[2, \infty)$ provided $u_b - \varepsilon \geq \log 2$ for every bump and $1 - \sum_m a_m x^{-m} \geq 0$ on (X, ∞) ; since $a_m \geq 0$ the last function is increasing, so $x = X$ suffices. In Lemma 2.2 (with $r = 1$), μ is explicit:

$$\mu(t) = \rho(t) + \frac{1}{\pi} \sum_{j=0}^n C_j e^{v_j/2} E(t, v_j) - \frac{S(t)}{\pi} \sum_b m_b \cos(tu_b) + \sum_m \frac{a_m}{\pi} X^{1/2-m} T_m(t),$$

with $E(t, v) = \operatorname{Re}[e^{itv}/(\frac{1}{2} + it)]$, $T_m(t) = \operatorname{Re}[e^{-itL}/(m - \frac{1}{2} + it)]$, $L = \log X$, $C_0 = 1 - f_0$, $C_j = f_{j-1} - f_j$ ($0 < j < n$), $C_n = f_{n-1}$ and $m_b = w_b e^{-u_b/2}$. The verifier proves $\mu \geq 0$ on a finite interval $[0, T]$ by mean-value steps, $\mu(t) \geq \mu(t_0) - (h + d) \sup |\mu'|$ on $[t_0 - d, t_0 + h + d]$, with μ' enclosed on the whole step in ball arithmetic. Beyond T it uses monotone majorants: ρ is increasing, while $|E(t, v)| \leq (\frac{1}{4} + t^2)^{-1/2}$, $|S(t)| \leq \min(1, (k/\varepsilon t)^k)$ and $|T_m(t)| \leq ((m - \frac{1}{2})^2 + t^2)^{-1/2}$ are decreasing, so for $t \geq T$, $\mu(t)$ is at least $\rho(T)$ minus the sum of these majorants at T , with the coefficients in absolute value.

Gaussian–Laguerre certificates. Let $f_k(u) := L_k^{(-1/2)}(2\pi u^2) e^{-\pi u^2}$; these are even Hermite functions, with $\widehat{f_k} = (-1)^k f_k$, the basis with prescribed double roots used for sign-uncertainty problems in [1]. The dual certificates below are functions

$$F_{\text{rep}}(t) := \sum_{k \leq K} a_k f_k(t/s) + \varepsilon (e^{-\pi t^2/(16s^2)} + e^{-16\pi t^2/s^2})$$

with explicit rationals a_k . Writing $F := \sum_k a_k f_k(t/s)$, $X = 2\pi t^2/s^2$ and $Y = 2\pi s^2 \xi^2$, one has $F = e^{-X/2} P(X)$ and $\widehat{F} = s e^{-Y/2} Q(Y)$ with $P, Q \in \mathbb{Q}[X]$, and exact Sturm sequences decide

the positivity of P on $(0, \infty)$ and of Q on the range of Y in question. The two Gaussians of the repair term are positive with positive Fourier transforms, so $F_{\text{rep}} \geq F$ and $\widehat{F}_{\text{rep}} \geq \widehat{F}$ pointwise; the repair term is not needed for these positivity statements, and it adds the explicit lower bounds $F_{\text{rep}}(t) \geq \varepsilon e^{-\pi t^2/(16s^2)}$ and, where $\widehat{F} \geq 0$, $\widehat{F}_{\text{rep}}(\xi) \geq \frac{\varepsilon s}{4} e^{-\pi s^2 \xi^2/16}$. $\mathcal{A}'(F_{\text{rep}})$ is one rigorous ball-arithmetic integral of F_{rep} against the digamma weight on a finite interval, plus an explicit tail bound.

Fact 2.3 (Results of [4] on ζ , $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{-3})$). (a) *For ζ 's data, $-2.7112 \cdot 10^{-3} \leq \kappa^* \leq 1.1508391 \cdot 10^{-906}$ ([4, Theorem C]); and $\kappa^* \leq \kappa_{\text{OPS}}^* \leq 9.9462 \cdot 10^{-41}$, certified by a Gaussian–Laguerre certificate with $K = 128$, $s = 13$, $\varepsilon = 10^{-56}$ whose transform is positive on all of $\mathbb{R}$ ([4, Proposition 5.3]).* (b) *For the archimedean data of $\mathbb{Q}(\sqrt{5})$ ($\Gamma_{\mathbb{R}}^2$, $q = 5$, $r = 1$) and every gap $g \in (0, \xi_{4.04915}]$, in particular for ξ_2 and for the natural gap ξ_4 of the field, $1.64 \cdot 10^{-5} \leq \kappa_g^* \leq 4.83 \cdot 10^{-4}$; for the data of $\mathbb{Q}(\sqrt{-3})$ ($\Gamma_{\mathbb{C}}$, $q = 3$, $r = 1$) and every gap $g \in (0, \xi_{3.011664}]$, in particular for ξ_2 and for the natural gap ξ_3 , $1.50 \cdot 10^{-4} \leq \kappa_g^* \leq 9.99 \cdot 10^{-4}$. In both cases some admissible pair has zero measure bounded below by a positive multiple of dt , and the set of admissible pairs is not a singleton ([4, Proposition 5.9]).*

The proof of (b) in [4] combines the lower bound of Theorem 3.1 below (resp. Proposition 3.2), read at $q = 5$ (resp. $q = 3$), with the explicit pair of the same statement. Corollary 3.3 below states the same certificates in terms of the critical conductors $q_{\min, g}$.

3. BRACKETS FOR TWO CRITICAL CONDUCTORS

Theorem 3.1 (A non-arithmetic critical point, bracketed). *For the data $\Gamma_{\mathbb{R}}^2$ with one simple pole ($\kappa = (0, 0)$, $r = 1$),*

$$4.98485094226764 \leq q_{\min} \leq 4.9994844.$$

More precisely: an explicit Gaussian–Laguerre certificate $F_{\text{rep}} \in \mathcal{C}$ with $K = 80$ and $s = 7$ has $\mathcal{A}_{(0,0),1,1}(F_{\text{rep}})/\int F_{\text{rep}} \leq -0.25566705789831547$, so no admissible pair exists for $q < 4.98485094226764$; and an explicit pair of the form above at $q_0 = e^{1.6093347792651136} \leq 4.9994844$ has $\mu \geq 6.59 \cdot 10^{-9}$ on $[0, 800]$ and $\mu \geq 1.79$ on $[800, \infty)$.

Proof (computer-assisted). Lower bound: with the notation above, exact Sturm sequences show that P has no root in $(0, \infty)$, that Q has no root in (Y_{10}, ∞) for a rational Y_{10} below the image of ξ_2 , and that the relevant end values and leading coefficients are positive; so $F_{\text{rep}} > 0$ on $\mathbb{R}$ and $\widehat{F}_{\text{rep}} > 0$ on $[\xi_2, \infty)$, and $F_{\text{rep}} \in \mathcal{C}$. In fact Q has no root in $(0, \infty)$ and $Q(0) > 0$, so $\widehat{F}_{\text{rep}} > 0$ on all of $\mathbb{R}$ and $F_{\text{rep}} \in \mathcal{C}_{\text{OPS}}$. The functional is one Arb integral at 448 bits (`python general/verify_general.py general/params/gammaR2_K80_s7.json`). Then $\mathcal{A}_{(0,0),q,1}(F_{\text{rep}}) < 0$ for $\log q < 2\pi \cdot 0.25566705789831547$, and Lemma 2.1 applies. Upper bound: `python general/verify_pair.py general/params/pair_gammaR2_X240.json` checks $\tilde{\nu} \geq 0$ and $\text{supp } \tilde{\nu} \subseteq [2, \infty)$ (the first edge is exactly $\log 2$ in the $\log x$ variable), and proves $\mu > 0$ on $[0, 800]$ by 17 214 mean-value steps with μ' enclosed in Arb (128 bits) on each whole step, and on $[800, \infty)$ by the monotone majorants described in Section 2. Lemma 2.2 gives admissibility at q_0 , and feasibility for $q \geq q_0$ follows by monotonicity. $\square$

Proposition 3.2 (The critical conductor of $\Gamma_{\mathbb{C}}$). *For the data $\Gamma_{\mathbb{C}}$ with one simple pole ($\kappa = (0, 1)$, $r = 1$), $2.981236704765 \leq q_{\min} \leq 2.997174$.*

Proof (computer-assisted). As for Theorem 3.1, with a Gaussian–Laguerre certificate for $\kappa = (0, 1)$ ($K = 64$, functions $f_k(t/9)$; `general/params/gammaC_K64_s9.json`, which prints $q_{\min} \geq 2.9812367047650810559$) and an explicit pair of the form above at $\log q_0 = 1.0976698108720329$ (`general/params/pair_gammaC_X40.json`), whose zero density satisfies the certified floor $\mu \geq$

$6.67 \cdot 10^{-8}$ on $\mathbb{R}$. As in Theorem 3.1, the transform of the dual certificate is positive on all of $\mathbb{R}$. $\square$

Neither bracket uses the value of the gap in an essential way: the dual certificates have transforms positive on all of $\mathbb{R}$, and the prime measures of the explicit pairs start well beyond 2.

Corollary 3.3 (The brackets at other gaps). (a) *For the data $\Gamma_{\mathbb{R}}^2$ with one simple pole, $4.98485094226764 \leq q_{\min,g} \leq 4.9994844$ for every gap g with $0 < g \leq \xi_{4.04915}$; and the data of $\mathbb{Q}(\sqrt{5})$ ($\Gamma_{\mathbb{R}}^2$, $q = 5$, $r = 1$) have $1.64 \cdot 10^{-5} \leq \kappa_g^* \leq 4.83 \cdot 10^{-4}$ for these g . (b) *For the data $\Gamma_{\mathbb{C}}$ with one simple pole, $2.981236704765 \leq q_{\min,g} \leq 2.997174$ for every gap g with $0 < g \leq \xi_{3.01166}$; and the data of $\mathbb{Q}(\sqrt{-3})$ ($\Gamma_{\mathbb{C}}$, $q = 3$, $r = 1$) have $1.50 \cdot 10^{-4} \leq \kappa_g^* \leq 9.99 \cdot 10^{-4}$ for these g . In particular, at the natural gaps ξ_4 of $\mathbb{Q}(\sqrt{5})$ and ξ_3 of $\mathbb{Q}(\sqrt{-3})$, some admissible pair has zero measure bounded below by a positive multiple of dt , and these data are not critical.**

Proof (computer-assisted). At a gap g , weak duality takes the form of Lemma 2.1(a) with C_g in place of C : for an admissible pair at the gap g and $F \in C_g$, both integrands in $\mathcal{A}'(F) = \int F d\mu + \frac{1}{\pi} \int \hat{F} d\nu$ are non-negative on the supports of the measures. The identity of Lemma 2.2 does not involve the gap, so its pair is admissible at the gap g if and only if $\mu \geq 0$ and $\tilde{\nu}$ is carried by $[e^{2\pi g}, \infty)$.

(a) The dual certificate of Theorem 3.1 lies in $C_{\text{OPS}} \subseteq C_g$ (its log reports $\text{F.rep}^{\wedge} > 0$ on $\mathbb{R}$ (OPS cone): `True`), so no admissible pair exists at any gap for $q < 4.98485094226764$, and at $q = 5$ it gives $\kappa_g^* \leq -0.25566705789831547 + \log 5 / 2\pi \leq 4.83 \cdot 10^{-4}$. Run with the option `--gap-x 4`, the verifier of the explicit pair of Theorem 3.1 proves, by exact comparisons of the cell edges and outward-rounded bounds for the bump supports, that `supp nu~ in [x0, oo) with x0 >= 4.049150`, and reports `admissible at gap xi_4 (prime measure on [4, oo))`: `True`. So the pair is admissible at every gap $g \leq \xi_{x_0}$, and $q_{\min,g} \leq 4.9994844$ there. With μ replaced by $\mu + \frac{1}{2\pi} \log(5/q_0)$ the same $\tilde{\nu}$ gives an admissible pair for the data of $\mathbb{Q}(\sqrt{5})$ whose zero measure is at least $1.64 \cdot 10^{-5} dt$, so $\kappa_g^* \geq 1.64 \cdot 10^{-5}$. The script `general/corollary_slack.py` combines the two runs: `python general/corollary_slack.py general/params/corollary_qsqrt5.json --gap-x 4` prints $4.984850942267643 \leq q_{\min} \leq 4.999484361$ and $1.6420748 \cdot 10^{-5} \leq \kappa^* \leq 0.00048294147$, and that both hold for every gap g with $e^{2\pi g} \leq x_0$.

(b) The same, with Proposition 3.2: its dual certificate lies in C_{OPS} ; the verifier of its pair, run with `--gap-x 3`, proves $x_0 \geq 3.011664$ and admissibility at the gap ξ_3 ; and the run of `general/corollary_slack.py` with `general/params/corollary_qsqrtm3.json --gap-x 3` prints $0.00015006672 \leq \kappa^* \leq 0.00099854968$ for the data of $\mathbb{Q}(\sqrt{-3})$. $\square$

Numerical observation 3.4 (The slack as a function of the gap). The slack κ_g^* at $g = \xi_n$ was also computed by exact sums-of-squares linear programmes in the Gaussian–Laguerre basis ($K \leq 80$, an interior-point solver at 640 bits, duality gaps below 10^{-24} ; every solution with $K \geq 32$ was re-checked at 256 bits on fine grids, with refined local minima); these computations are not part of the ancillary files. For ζ 's data and for those of $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{-3})$, κ_g^* is constant for $0 < g \leq \xi_{n_1}$, where n_1 is the first prime atom of the extremal pair of the linear programme, and equals there the slack κ_{OPS}^* of the cone C_{OPS} ; beyond ξ_{n_1} it decreases roughly linearly in g , with slope between -1.85 and -1.5 , and becomes negative. For ζ 's data the computed values of the constant decay like $10^{-0.3K}$, consistent with 0 (unconditionally only $-2.7112 \cdot 10^{-3} \leq \kappa^* \leq 1.1508391 \cdot 10^{-906}$ is known, and $\kappa^* \geq 0$ holds if admissible pairs exist, for instance under the Riemann hypothesis), and $n_1 = 2$; the slack is -0.0470 at $\xi_{2.5}$ and -0.0793 at ξ_3 . For the data of $\mathbb{Q}(\sqrt{5})$ the constant is $4.8294 \cdot 10^{-4}$ and $n_1 \approx 4.3201$; the slack vanishes near $n \approx 4.328$ and is -0.0372 at ξ_5 . For the data of $\mathbb{Q}(\sqrt{-3})$ the constant is $9.9855 \cdot 10^{-4}$

and $n_1 \approx 3.2442$; the slack vanishes near $n \approx 3.255$ and is -0.0570 at ξ_4 . Where Corollary 3.3 applies, every computed value with $K \geq 32$ lies in its bracket. In these examples the extremal pair for ζ 's data is arithmetic, its first prime atom being the prime 2, while for the two fields the first atom falls between prime-ideal norms and the constant is positive; we draw no general conclusion from this.

Remark 3.5 (Comparison with Odlyzko's tables). If the generalised Riemann hypothesis holds for the Dedekind zeta function of a real (resp. imaginary) quadratic field k , then its own pair of measures (zeros on the zero side, prime ideals and their powers on the prime side, which lies beyond ξ_2 because every norm is at least 2) is admissible for the data $\Gamma_{\mathbb{R}}^2$ (resp. $\Gamma_{\mathbb{C}}$) at $q = |d_k|$, $r = 1$, by the explicit formula for the Dedekind zeta function of k , which holds on $\mathcal{T}$ by the argument used for ζ ; so $|d_k| \geq q_{\min}$. In the normalisation of root discriminants, the lower ends of Theorem 3.1 and Proposition 3.2 are $4.98485^{1/2} = 2.2327$ and $2.98123^{1/2} = 1.7266$, against the GRH bounds 2.225 and 1.722 in degree 2 of [7, Tables 4 and 3, p. 134] (see also [6, 3]); the minimal root discriminants are $5^{1/2} = 2.236$ and $3^{1/2} = 1.732$. Squared, the tabulated bounds are $2.225^2 \approx 4.95 < 5$ and $1.722^2 \approx 2.97 < 3$, consistent with our brackets. In the sense of Cohn and Triantafyllou [2], the explicit pairs of Theorem 3.1 and Proposition 3.2 are dual certificates of non-sharpness: no test function in $\mathcal{C}$ can prove a conductor bound above 4.9994844, resp. 2.997174, so for these data the method cannot reach the minimal discriminants 5 and 3.

Remark 3.6 (The role of the gap). Except in Corollary 3.3, this note uses the gap $\xi_2 = \log 2/2\pi$, the first prime of $\mathbb{Q}$, for every archimedean type. For the data of a number field $k \neq \mathbb{Q}$ this gap need not be natural. In $\mathbb{Q}(\sqrt{5})$ the primes 2 and 3 are inert, so the norms of the prime ideals are 4, 5, 9, 11, 11, $\dots$; in $\mathbb{Q}(\sqrt{-3})$ the first norm is 3. At the gap ξ_4 , resp. ξ_3 , positivity of $\hat{F}$ is required only beyond that point, so the cone is larger and κ^* can only decrease; and if the generalised Riemann hypothesis holds for ζ_k , the field's own pair is admissible at that gap, so there $\kappa^* \geq 0$. So positive slack at the gap ξ_2 does not by itself give positive slack at the natural gap. For these two fields it does (Corollary 3.3): the dual certificates have transforms positive on all of $\mathbb{R}$, and the explicit pairs put no prime mass below $x = 4.049$, resp. $x = 3.011$. The margins are small: the critical conductor of the data $\Gamma_{\mathbb{R}}^2$ with one simple pole, which are those of $\mathbb{Q}(\sqrt{5})$ up to the conductor, lies in $[4.9848, 4.9995]$ at every gap up to $\xi_{4.049}$, so its distance below the discriminant 5 is between 10^{-4} and about $3 \cdot 10^{-3}$ in relative terms. For the other types of Numerical observation 5.1 the computations were made at the gap ξ_2 only, and their comparison with minimal discriminants may depend on the gap. In general the critical conductor can depend on the gap, and so can Question 6.1.

Proposition 3.7 (An admissible pair for the data of $\mathbb{Q}(i)$). *For the data of $\mathbb{Q}(i)$ ($\Gamma_{\mathbb{C}}$, $q = 4$, $r = 1$) there is an admissible pair of the form above with $\mu \geq 0.0217$ on $[0, 300]$ and $\mu \geq 0.4696$ on $[300, \infty)$. Hence $\kappa^* \geq 0.0217$ for these data, and the set of admissible pairs is not a singleton.*

Proof (computer-assisted). As for the upper bound in Theorem 3.1: the script `general/verify_pair.py` with `general/params/pair_Qi_X10.json` checks $\tilde{\nu} \geq 0$ with support in $[2, \infty)$ and proves the bounds for μ by 1277 mean-value steps with μ' enclosed in Arb, and by monotone majorants beyond 300; the printed floor is $\mu \geq 0.0222324$ on $\mathbb{R}$. Lemma 2.2 gives admissibility, and $\mu \geq 0.0217$ dt gives $\kappa^* \geq 0.0217$ by Lemma 2.1. Adding to $\tilde{\nu}$ a small atom $w\delta_{x_1}$, $x_1 \geq 2$, changes μ by $-(w/\pi)x_1^{-1/2}\cos(t\log x_1)$; for small $w > 0$ the result is a second admissible pair. $\square$

4. THE POLE WEIGHT OF ζ 'S DATA

Theorem 4.1 (The pole weight is pinned). *For ζ 's data with pole weight r , that is $\mathcal{A}_r(F) := r(F(\frac{i}{2}) + F(-\frac{i}{2})) + \int F\Omega_{\infty}$ with $\Omega_{\infty}(t) = \frac{1}{2\pi}(\operatorname{Re}\psi(\frac{1}{4} + \frac{it}{2}) - \log \pi)$, there is no admissible pair*

if $r < 1 - 5.5794 \cdot 10^{-40}$ or $r > 1 + 7.8550942 \cdot 10^{-21}$. Under the Riemann hypothesis, $r = 1$ is feasible.

Proof (computer-assisted). $\mathcal{A}_r(F) = \mathcal{A}_1(F) + (r-1)D$ with $D = F(\frac{i}{2}) + F(-\frac{i}{2})$, so one certificate $F \in \mathcal{C}$ per side suffices. For the left side, the $K = 128$ certificate of Fact 2.3(a) has $D > 0$, and its run (`python general/verify_general.py general/params/zeta_K128_s13.json`, also in the ancillary files of [4]) prints $\mathcal{A}_1(F)/D \leq 5.57930810147 \cdot 10^{-40}$. For the right side, a certificate with $K = 80$, $s = 12$, $\varepsilon = 10^{-30}$, whose transform is positive on $[\xi_2, \infty)$ but negative on part of $(0, \xi_2)$, has $D < 0$ and $\mathcal{A}_1(F)/|D| \leq 7.85509419726 \cdot 10^{-21}$ (`python general/verify_general.py general/params/zeta_K80_s12_edge.json`). In both cases $\mathcal{A}_r(F) < 0$ beyond the stated thresholds, and Lemma 2.1 applies. Under the Riemann hypothesis, ζ 's own pair p_ζ is admissible for $r = 1$ [4]. $\square$

The same verifier certifies $\kappa_{\text{OPS}}^* \leq 7.24651 \cdot 10^{-32}$ with $K = 104$ (parameter file `general/params/zeta_K104_s13.json`); this certificate also has $D > 0$, with $\mathcal{A}_1(F)/D \leq 4.06492480958 \cdot 10^{-31}$, which gives the weaker left pin $r < 1 - 4.0649249 \cdot 10^{-31}$. Like the bound $\kappa_{\text{OPS}}^* \leq 9.9462 \cdot 10^{-41}$ for $K = 128$, the bound for κ_{OPS}^* needs no gap at ξ_2 (the transform of the certificate is positive on all of $\mathbb{R}$), so it is directly comparable with the classical tables [7, Table 4]. Only this pinning is certified; we make no claim about the shape of the feasible region in other directions.

5. NUMERICAL OBSERVATIONS

The observations below were computed but not certified, except where a certified statement is cited; none is used in a proof.

Numerical observation 5.1 (Critical conductors of other Gamma types). For each Gamma type with one simple pole, the critical conductor $e^{-2\pi\kappa_0^*}$, where κ_0^* is the slack at $q = 1$, was computed by linear programming at finite K (converged in K , and reproduced to 8 digits of κ_0^* by an independent implementation; not certified except as in Theorem 3.1 and Proposition 3.2). We list it against the smallest absolute discriminant of a number field with these archimedean data: $\Gamma_{\mathbb{C}}$: 2.981 / 3; $\Gamma_{\mathbb{R}}$: 4.985 / 5; $\Gamma_{\mathbb{R}}\Gamma_{\mathbb{C}}$: 22.6 / 23; $\Gamma_{\mathbb{R}}^3$: 48.2 / 49; $\Gamma_{\mathbb{C}}^2$: 114.7 / 117; $\Gamma_{\mathbb{R}}^2\Gamma_{\mathbb{C}}$: 267.8 / 275; $\Gamma_{\mathbb{R}}^4$: 694.6 / 725. With the gap ξ_2 , in every case tested the genuine Dedekind zeta function lies strictly above its critical conductor. For $\Gamma_{\mathbb{R}}^2$ and $\Gamma_{\mathbb{C}}$ this also holds at the natural gaps of $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{-3})$, and is certified (Corollary 3.3); for the other five types the computations were made at the gap ξ_2 only, and the comparison may depend on the gap (Remark 3.6). For ζ , by contrast, $1 - q_{\min} \leq 7.3 \cdot 10^{-906}$ (Fact 2.3(a)), and $q_{\min} \leq 1$ under the Riemann hypothesis.

Numerical observation 5.2 (The near-critical objects are not arithmetic). Read from the parameter files: below $x \approx 17.8$ the prime measure of the explicit pair of Theorem 3.1 is a sum of narrow bumps (half-width 0.05 in $\log x$), the first ones centred near $x \approx 4.257, 4.277, 8.410, 8.591, 11.57$ and 12.47 ; and the prime measure of the pair of Proposition 3.7 has atoms at $x \approx 3.089, 3.113, 4.693, 4.731, 5.425$ and 5.469 , and so on. The computed critical object for the data $\Gamma_{\mathbb{R}}^2$ (from the linear programmes of Numerical observation 3.4) has zeros near 6.75, 10.44, 12.83 and prime atoms near $x \approx 4.3201, 9.215, 14.41$, and so on. None of these positions is an integer, so none of these measures resembles the prime-ideal measure of a number field (respectively of $\mathbb{Z}[i]$). So the critical object for these data appears to be non-arithmetic; this reading is heuristic.

Numerical observation 5.3 (Window masses near the critical point of $\Gamma_{\mathbb{R}}^2$). For the data $\Gamma_{\mathbb{R}}^2$, the largest mass that an admissible pair can put in four windows, two on the prime side (the images of $x \in (2.05, 3.95)$ and of $x \in (5, 8.5)$) and two on the zero side ($(0.5, 6)$ and $(7.6, 9.7)$), was computed by finite linear programmes ($K = 48$) at conductors $\log q = \log q_{\min} + \delta$. It decreases as $\delta \downarrow 0$, from 0.314, 1.26, 0.078 and 0.67 at $\delta = 10^{-3}$ to 0.0044, 0.148, 0.00085 and 0.022 at

$\delta = 10^{-5}$, while for the slack data of $\mathbb{Q}(\sqrt{5})$ the same windows carry up to 0.60, 1.57, 0.20 and 0.94. These computations are not reproduced in the ancillary files. They are consistent with a single admissible pair at the critical conductor, but these computations at $\delta > 0$ do not show this: that would need a convergence statement for the dual problems as $\delta \rightarrow 0$, which has only been computed. Similarly, in three one-parameter families of Gamma factors, finite linear programmes ($K = 48$) showed no visible fattening of the set of admissible pairs at the critical conductor; finite computations of this kind can suggest, but not show, that a critical fibre is a single pair.

6. A QUESTION

For other archimedean data the question of uniqueness at criticality is open even numerically.

Question 6.1 (Uniqueness at criticality for other Gamma factors). *Let $\gamma = \Gamma_{\mathbb{R}}^a \Gamma_{\mathbb{C}}^b$ be a Gamma factor, with one simple pole at $s = 1$ and the gap ξ_2 , and let $q_c(\gamma)$ be its critical conductor, at which the slack vanishes. Is the set of admissible pairs at $q = q_c(\gamma)$ a single pair?*

The critical conductor, and with it the question, can depend on the gap (Remark 3.6). For ζ 's own type the question concerns Conjecture U of [4] only when $\kappa^* = 0$, for instance under the Riemann hypothesis and Conjecture S of [4]. Then $q_c = 1$, admissible pairs exist, and the question asks whether there is only one; under the Riemann hypothesis, which makes ζ 's own pair admissible, this is Conjecture U. If $\kappa^* \neq 0$, it concerns a conductor $q_c \neq 1$ and is unrelated to Conjecture U. For $\Gamma_{\mathbb{R}}^2$ the critical conductor is bracketed by Theorem 3.1. A first step would be to certify, at the critical conductor itself or at certified conductors converging to it, upper bounds for the window masses of Numerical observation 5.3.

AI-USE STATEMENT

This note was produced with substantial use of AI. The mathematical arguments, the numerical computations and the text were generated by instances of Claude Opus 5.5 (Anthropic) working as coordinated agents: one set of agents developed the arguments, separate agents acted as independent hostile referees of each component, and the referees' requested repairs were incorporated. The numerical certificates were computed by AI-written scripts that are included as supplementary material and can be rerun. The author initiated and directed this project, designed its verification process, and reviewed the results and the verification record, but did not independently check every step of the written proofs. The author takes full responsibility for the content of this note, including any errors, and welcomes independent verification and corrections.

APPENDIX A. ANCILLARY FILES

Every computer-assisted statement names its script and parameter file; the scripts need Python ≥ 3.10 with python-flint 0.9.0 and mpmath 1.3.0, and are run from the top directory of the ancillary files of this note, which are self-contained: the scripts `general/verify_general.py`, `general/verify_pair.py` and `general/corollary_slack.py`, the parameter files in `general/params/`, the summaries of the runs in `general/results/` and the logs in `general/logs/`. The certificates `zeta_K104_s13.json`, `zeta_K80_s12_edge.json` and `pair_Qi_X10.json` are used only here; the $K = 128$ certificate `zeta_K128_s13.json` and the certificates for $\Gamma_{\mathbb{R}}^2$ and $\Gamma_{\mathbb{C}}$ (`gammaR2_K80_s7.json`, `pair_gammaR2_X240.json`, `gammaC_K64_s9.json`, `pair_gammaC_X40.json`) are also shipped, as identical files, with the ancillary files of [4]. The script `general/run_all.sh` repeats all of them (about 25 CPU-minutes), and `sha256sum -c`

SHA256SUMS checks the inputs. The ancillary files are licensed under the Apache License 2.0 (copyright 2026 Nic Johns); the text is licensed under CC BY 4.0.

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